



FLEET & SHORE PERSONNEL TRAINING

Senior Engine Officers Adaptation & Refreshment Training

A structured refresher for Chief and Second Engineers joining a Glory-managed vessel, covering machinery management, emergency response, environmental compliance and engine-room leadership.

STCW Reg. III/2

ISM Code

MARPOL Annexes I & VI

Company SMS

What We Will Cover

01

The Senior Engineer's Role

Authority, accountability and the STCW competence framework

02

Machinery Management & PMS

Planned maintenance, spares and critical-equipment oversight

03

Engine-Room Emergency Procedures

Fire, flooding, blackout and machinery-space casualties

04

MARPOL & Environmental Compliance

Oily-water, sludge, emissions and record-keeping oversight

05

Leadership & Mentoring

Running a safe, disciplined and well-trained engine department

06

SMS Refresher

Company procedures, permits and the engineer's reporting duties



SECTION 1

The Senior Engineer's Role

What changes — and what does not — when an experienced engineer joins a new vessel or returns to sea after time ashore.

01

Why Adaptation & Refreshment Matters

Senior engine officers carry a level of technical and legal responsibility that demands a deliberate familiarisation process on every new assignment.

- A change of vessel, engine type or trading pattern is one of the highest-risk moments in an engineer's career — most machinery-space incidents cluster in the first weeks aboard.
- This training refreshes core competence, introduces vessel-specific systems, and confirms the officer understands the Company's expectations before taking charge of the watch.
- It applies equally to a Chief Engineer joining a new ship and to a Second Engineer returning from leave or promoted into the role.
- STCW Regulation III/2 and Section A-III/2 set the underpinning competence standard; this course maps directly onto that framework for management-level engineers.
- Completion is recorded in the officer's training file and forms part of the vessel's ISM familiarisation record.



Familiarisation First

No senior engineer should assume sole charge of the watch until vessel-specific familiarisation — machinery layout, critical valves, emergency shutdowns — is complete and signed off.

AREAS OF RESPONSIBILITY

What the Chief Engineer Answers For

The Chief Engineer holds overall responsibility for the technical operation and safety of the engine department, under the Master's overall command.



Machinery Safety & Availability

Keeping propulsion, steering, power generation and essential auxiliaries safe, reliable and ready for the voyage ahead.



Statutory & Class Compliance

Ensuring machinery, certificates and surveys meet flag-state, class and MARPOL requirements at all times.



Department Management

Watchkeeping rosters, work planning, spares and budget control across the engine team.



Crew Competence & Welfare

Training, mentoring and monitoring the fitness for duty of engine officers and ratings.

First Days Aboard

A disciplined handover and familiarisation routine sets the tone for the whole assignment.



Do

- Complete a full handover with the outgoing engineer, in writing where practicable
- Walk the engine room and machinery spaces before taking sole charge of the watch
- Review outstanding defects, PMS status and critical spares on arrival
- Confirm familiarisation with emergency shutdowns, isolations and escape routes



Don't

- Don't accept sole charge of the watch before familiarisation is complete
- Don't assume vessel systems match your previous ship without checking
- Don't sign off a handover you have not personally verified
- Don't delay reporting concerns about the vessel's condition to the Master or Company



SECTION 2

Machinery Management & PMS

02

Running the planned maintenance system as a management tool, not a paperwork exercise.

Owning the PMS

The Company's Planned Maintenance System exists to keep machinery reliable and to provide objective evidence of that reliability.

- The Chief Engineer is accountable for PMS compliance across the department, even where individual jobs are delegated to other officers or ratings.
- Running-hours-based and calendar-based jobs must both be tracked; overdue jobs should be reviewed personally, not simply rescheduled without cause.
- Critical machinery (main engine, steering gear, emergency generator, fire pumps) requires closer, documented oversight than routine auxiliary equipment.
- Deferred maintenance must be risk-assessed and, where it affects safety-critical equipment, reported to the Company through the SMS.
- PMS records are evidence during PSC inspections, vetting inspections and internal audits — accuracy matters as much as completion.



Trend, Don't Just Record

Reviewing running parameters and condition-monitoring data over time catches deterioration long before a job falls due on the PMS calendar.

CRITICAL EQUIPMENT

Oversight by Equipment Category

Not all machinery carries the same risk — oversight effort should be weighted accordingly.

Category	Examples	Senior Officer's Focus
Propulsion & Steering	Main engine, steering gear, controllable-pitch propeller	Daily condition checks; immediate reporting of any irregularity
Critical Safety Equipment	Emergency generator, fire pumps, bilge system, watertight doors	Scheduled testing per SMS; verified, not just recorded, as tested
Power Generation	Auxiliary engines, switchboard, automation	Load management, blackout prevention, spares availability
Routine Auxiliaries	Purifiers, compressors, pumps	Delegated to junior officers with periodic senior review

Spares, Budget and Dockyard Planning

Effective machinery management extends beyond the engine room to resource planning ashore.



Critical Spares Inventory

Verifying minimum spares required by class and Company policy are aboard, in date and correctly stored.



Requisitioning

Timely, justified spares and consumables requests that anticipate the voyage plan rather than react to failures.



Drydock & Repair Planning

Building and prioritising the repair list, and briefing the Company Technical Department well ahead of yard periods.



Condition-Based Reporting

Monthly technical reports that give shore management an accurate, honest picture of machinery condition.



SECTION 3

Engine-Room Emergency Procedures

03

Leading the engine team's response when seconds and correct decisions both matter.

The Chief Engineer's Emergency Duties

Every engine-room emergency scenario in the Company's Contingency Plan assigns the Chief Engineer a defined command role.

1

Direct the engine team's immediate response and confirm machinery status to the Bridge

2

Isolate the affected system — fuel, electrical or mechanical — without delay

3

Coordinate with the Master on propulsion and power availability throughout the incident

Machinery-Space Fire and Flooding

Engine-room fires and flooding remain among the most dangerous casualties a ship can suffer.

- On discovering fire, raise the alarm immediately, stop fuel supply to the affected area if safe to do so, and begin boundary cooling before committing to attack the fire.
- Quick-closing valves, emergency fuel shut-offs and ventilation dampers must be known by location and function, not just by name, before an emergency occurs.
- For flooding, identify and isolate the source where possible, start bilge pumping and monitor list and trim in close contact with the Bridge.
- Fixed fire-extinguishing systems (CO2 or equivalent) are released only on the Master's order, after a confirmed muster and boundary check — never automatically.
- Every drill is an opportunity to test real response times against the SMS's target times, not simply to complete a checklist.



Muster Before Release

Before any fixed CO2 system is released, a headcount must confirm no one remains in the space and all boundaries and ventilation are secured.

Blackout Recovery

A blackout at the wrong moment — in restricted waters, in heavy weather — can escalate rapidly without disciplined recovery.



Do

- Immediately inform the Bridge of the blackout and estimated recovery time
- Follow the vessel's black-start procedure in the correct sequence
- Confirm steering gear and essential services are restored before full manoeuvring
- Investigate and record the root cause once power is safely restored



Don't

- Don't attempt shortcuts in the black-start sequence under pressure
- Don't restore non-essential loads before propulsion and steering are secure
- Don't assume a single restart attempt without checking the trip cause
- Don't delay a Company incident report for a blackout in restricted waters



SECTION 4

MARPOL & Environmental Compliance

The senior engineer's personal accountability for the ship's environmental record.

04

Why This Sits With the Chief Engineer

MARPOL violations are among the most heavily penalised offences in shipping, and prosecutors target the Chief Engineer directly, not only the Company.

- The Chief Engineer is personally responsible for the accuracy of the Oil Record Book, Garbage Record Book and related statutory logs.
- Any bypass, illegal discharge or falsified entry exposes the individual officer to criminal liability, detention and loss of certificate, independent of Company policy.
- MARPOL Annex I (oil), Annex V (garbage) and Annex VI (air emissions, sulphur limits) all fall within the engine department's direct operational control.
- The Company's zero-tolerance policy on illegal discharge and record falsification applies without exception, regardless of commercial pressure or time constraints.
- Environmental compliance is verified not only by flag-state and PSC inspection but by the Company's own internal and external ISM audits.



When in Doubt, Report

An engineer who discovers a genuine equipment failure affecting discharge compliance must report it to the Master and Company immediately — an honest report is protected; a concealed failure is not.

Key Environmental Systems

Each system requires the senior engineer's direct verification that it is operated correctly, not delegated blindly.



Oily-Water Separator (OWS)

Verified operational, sealed and free of any unauthorised bypass line before every use; alarms tested per schedule.



Sludge & Bilge Management

Sludge tank capacity and disposal planned well ahead of port calls to avoid pressure to discharge illegally.



Ballast Water Treatment System (BWTS)

Operated and logged per the vessel's Ballast Water Management Plan on every exchange or treatment cycle.



Fuel Sulphur Compliance

Bunker delivery notes, changeover logs and fuel sampling verified against MARPOL Annex VI sulphur limits.

Statutory Logs the Chief Engineer Verifies

Correct, contemporaneous record-keeping is the primary defence in any environmental inspection.

Record	Governing Annex	Verification Point
Oil Record Book (Part I)	MARPOL Annex I	Signed same day as the operation; matches actual tank soundings
Garbage Record Book	MARPOL Annex V	All categories logged; disposal port/facility recorded
Bunker Delivery Notes & Fuel Log	MARPOL Annex VI	Sulphur content matches ECA/global limits in force
Ballast Water Record Book	BWM Convention	Exchange/treatment position and volumes match the plan



SECTION 5

Leadership & Mentoring

Building a competent, disciplined and motivated engine department — not just running the machinery.

05

Leading the Engine Team

Engine Resource Management (ERM) applies the same human-factors principles as Bridge Resource Management to the engine control room and machinery spaces.

- Clear task allocation, closed-loop communication and cross-checking reduce the single-point failures that cause most machinery incidents.
- A senior engineer who accepts challenge from junior officers and ratings — and encourages them to speak up — builds a safer department than one who does not.
- Fatigue, workload and watch scheduling are the Chief Engineer's direct responsibility to manage, not simply the individual officer's problem.
- Toolbox talks before non-routine or hazardous jobs should be genuine risk discussions, not a formality to be signed and filed.
- A well-run engine department is measured by the competence of the people leaving it, not only by the condition of the machinery.



Mentoring Junior Officers

Deliberately involving cadets and junior engineers in fault diagnosis, PMS planning and permit-to-work preparation builds the Company's next generation of senior engineers.

Everyday Mentoring Habits

Mentoring works best woven into daily engine-room routine, not scheduled as a separate activity.



Explain the Why, Not Just the What

Talking through the reasoning behind a maintenance decision or an operational limit builds understanding that a bare instruction never does.



Debrief After Drills and Incidents

A short, blame-free debrief after every drill or near miss turns experience into lasting departmental knowledge.



Delegate With Oversight, Not Abdication

Hand real responsibility to junior officers while checking their work — competence grows fastest when mistakes are caught and corrected early.



SECTION 6

SMS Refresher

A practical review of the Company procedures every senior engineer is expected to apply without prompting.

06

Applying the SMS in the Engine Room

The Safety Management System is the practical backbone of everything covered in this course.

- Permit-to-work procedures apply to hot work, enclosed-space entry, working aloft and electrical isolation — no exceptions for time pressure or familiarity with the job.
- Management of Change procedures apply to any modification of machinery, safety equipment or operating parameters, however minor it may appear.
- Non-conformities and near misses must be reported through the SMS promptly and honestly — they are the Company's primary tool for preventing recurrence.
- Internal audits and Company inspections of the engine department should be treated as opportunities to verify and improve, not as exercises to pass.
- The Designated Person Ashore (DPA) is available at any time when an engineer believes safety or compliance is being compromised.



Document As You Go

Records completed at the time of the work are accurate and defensible; records reconstructed later from memory are neither.

KEY TAKEAWAYS

What Every Senior Engineer Should Leave Remembering

Three principles that apply across every topic covered in this course.

Verify

Personally check critical systems and records rather than relying solely on delegation

Report

Raise concerns, defects and non-conformities early, honestly and through the SMS

Develop

Build the competence of the whole engine team, not only your own technical skill



COURSE COMPLETE

A safe, compliant and well-run engine room starts with the example the senior engineer sets every single day.

Company DPA — 24/7 emergency line

Fleet Technical Department — mng@gloryship.com.tr

Fleet Safety Department — mng@gloryship.com.tr